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Compact Hole-Selective Self-Assembled Monolayers Enabled by Disassembling Micelles in Solution for Efficient Perovskite Solar Cells

*Ming Liu, Leyu Bi, Wenlin Jiang, Zixin Zeng, Sai-Wing Tsang, Francis R. Lin, and Alex K.-Y. Jen**

Self-assembled monolayers (SAMs) are widely employed as effective hole-selective layers (HSLs) in inverted perovskite solar cells (PSCs). However, most SAM molecules are amphiphilic in nature and tend to form micelles in the commonly used alcoholic processing solvents. This introduces an extra energetic barrier to disassemble the micelles during the binding of SAM molecules on the substrate surface, limiting the formation of a compact SAM. To alleviate this problem for achieving optimal SAM growth, a co-solvent strategy to disassemble the micelles of carbazole-based SAM molecules in the processing solution is developed. This effectively increases the critical micelle concentration to be above the processing concentration and enhances the reactivity of the phosphonic acid anchoring group to allow densely packed SAMs to be formed on indium tin oxide. Consequently, the PSCs derived from using MeO-2PACz, 2PACz, and CbzNaph SAM HSLs show universally improved performance, with the CbzNaph SAM-derived device achieving a champion efficiency of 24.98% and improved stability.

1. Introduction

Perovskite-based solar cells are very promising due to their low-temperature solution processibility and excellent optoelectronic properties.[1–3] The power conversion efficiency (PCE) of single-junction perovskite solar cells (PSCs) has already reached 26.1% which is comparable to that of crystalline silicon cells,[4] showing a bright future ahead for the commercialization of PSCs. Apart from engineering the perovskite compositions[5] and crystallization kinetics,[6–7] interfacial engineering has been employed as an effective tool in improving the efficiency and stability of PSCs.[8–10] Especially in inverted (p-i-n) PSCs, the hole-selective layer (HSL) between the perovskite absorber and indium tin oxide (ITO) anode plays a crucial role in influencing the device performance.[11–12] Recently, self-assembled monolayers (SAMs) have been applied as very effective HSLs to replace conventional organic hole-transporting materials such as PTAA and PEDOT:PSS, as SAMs benefit from their easily tuned energy level alignment with perovskite, minimized parasitic absorption, sufficient stability, and facile processibility that is compatible with low-cost and scalable production.[13–14] These SAM molecules usually consist of an anchor group that can bind with the substrates, a flexible spacer to provide sufficient rotational freedom for self-assembly, and a functional head group to facilitate efficient hole extraction.[15] Through rational molecular design, an ultra-thin and efficient HSL can be robustly formed on the ITO anode in inverted PSCs. The application of SAMs as an HSL also significantly reduces the charge-transport loss, leading to improved short-circuit current density (J_{SC}) and open-circuit voltage (V_{OC}) in PSCs for achieving record-high PCEs and impressive stability in single-junction,[16–17] tandem,[18] and flexible PSCs.[19]

A densely packed SAM with minimal defects is crucial for functioning as an efficient HSL in highly efficient and stable PSCs. Not only can SAMs prevent direct contact between the active layer and electrode, but they also enhance hole extraction to reduce interfacial charge carrier recombination.[20] Significant efforts have been spent to tailor the growth of SAM on the substrate. Getautis et al. reported a SAM named V1036 with a large head group as the HSL in inverted PSCs.[21] Albrecht et al. then reported two SAMs, 2PACz and MeO-2PACz, by removing the large diarylamine substituents from carbazole and replacing them with a pristine or dimethoxy-substituted carbazole head group. This significantly reduces the steric hindrance to result in tighter SAM packing, enabling a certified PCE of 23.26% to be realized in CIGS/perovskite tandem cells.[22]

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More recently, our group reported a CbzNaph SAM based on the helical π-extension of carbazole.[23] The resulting denser and more ordered packing of the CbzNaph SAM enables improved device efficiency and stability to be achieved.[24] Tang et al. also utilized a similar SAM HSL in wide-bandgap PSCs to facilitate the growth of high-quality perovskite and efficient hole extraction to realize a record-high PCE of 27.0% in 1 cm^2 monolithic all-perovskite tandem cells.[25] Aside from modifying the SAM functional head groups, the co-SAM strategy has also been employed to mitigate the defects on ITO and SAM layer. Getautis et al. mixed a butylphosphonic acid with V1036 to form a co-SAM with better coverage.[21] Our group also reported the incorporation of an alkyl ammonium SAM, IAHA, into a carbazole-based DC-PA SAM to realize a better aligned interfacial energy level and minimized SAM defects, resulting in a PCE of 23.5% with improved device stability.[26]

It is noteworthy that most of the hole-selecting SAM molecules are amphiphilic in nature,[27–29] therefore, these SAM molecules tend to form micelle nanoparticles in solution instead of staying dispersed as single molecules under commonly used processing concentrations (≈1 mg mL^{-1} in isopropanol (IPA)). Extra energy is needed to disassemble these micelles when they are deposited on the ITO substrate, leading to defect formation during SAM growth.[30] These defects often leave ITO exposed without proper SAM coverage, resulting in charge carrier recombination at the perovskite/HSL interface which reduces the efficiency and stability of PSCs. Furthermore, these micelles tend to aggregate with each other or even precipitate out from the solution, thereby severely reducing the shelf life of SAM inks for practical applications.

To better understand and mitigate these issues associated with formed SAM micelles, we have selected a set of examples based on 2PACz, MeO-2PACz, and CbzNaph to elucidate how micelle formation in SAM precursor solution affects the device performance of PSCs. The critical micelle concentration (CMC) of these SAMs in IPA was determined to be between 0.01 and 0.07 mg mL^{-1}, which is orders-of-magnitude lower than that in commonly used processing concentration (≈1 mg mL^{-1}).[31] To resolve this problem, *N,N*-dimethylformamide (DMF) was introduced into the SAM precursor IPA solution as a co-solvent. It was found through the dynamic light scattering (DLS) experiments that SAM micelles can be effectively disassembled due to stronger solvent-solute interaction between DMF and carbazole group.[32] As these micelles could be disassembled in the precursor solution, the extra energy needed for disrupting micelles during SAM growth can be mitigated. This facilitates a higher packing density and better coverage of SAM HSL on the ITO substrate which leads to better aligned energy levels, more efficient hole extraction, and reduced interfacial recombination loss.

Consequently, the V_{OC} and fill factor (FF) of devices can be universally improved by using co-solvent processed 2PACz, MeO-2PACz, and CbzNaph HSLs. An excellent PCE of 24.98% with an exceptionally high FF (85.06%) and good device stability could be achieved by employing the co-solvent processed CbzNaph (Cos-CbzNaph) HSL. This work shows the clear correlation between SAM micelle formation in the precursor solution and the final PSC performance, demonstrating a simple, effective, and generally applicable way to realize densely packed SAM HSLs for highly efficient and stable PSCs.

2. Results and Discussion

The self-assembling properties of SAM molecules in IPA were first evaluated by DLS measurements. The CbzNaph, MeO-2PACz, and 2PACz SAM molecules are intrinsically amphiphiles, possessing a hydrophobic carbazole head group and a hydrophilic phosphonic acid anchoring group on the same molecule (Figure 1a). When dispersed into IPA above the CMC, these SAM molecules tend to self-organize into micellar structures spontaneously; the non-polar carbazole-based groups face inwardly to the core and the polar phosphonic acids become exposed outwardly to the polar alcoholic environment (Figure 1b).[33] To determine the CMC, which is defined as the lowest concentration for forming SAM micelles, DLS measurements were performed on solutions to assess the hydrodynamic radius in the concentrations between 0.005 and 0.2 mg mL^{-1} (Figure 1c–e; Figure S1, Supporting Information). A sudden increase in hydrodynamic radius indicates the formation of micelles.[34] The CMCs of CbzNaph, MeO-2PACz, and 2PACz were determined as 0.018, 0.044, and 0.067 mg mL^{-1}, respectively, in IPA, which are orders-of-magnitude lower than those of commonly used processing concentrations (0.3–1.5 mg mL^{-1}). In addition, we confirmed the CMCs of SAMs by observing the sudden increase of scattered light intensity due to the formation of micelles.[35] The CMCs of CbzNaph, MeO-2PACz, and 2PACz were further determined as 0.013, 0.040, and 0.064 mg mL^{-1} (Figure S2, Supporting Information).

When SAMs are processed by spin-coating from their IPA solutions with low CMC, the relatively low specific area of micelles will limit the interactions between the micelle surface and substrate, thereby reducing the reactivity between the phosphonic acid anchoring group and the hydroxyl-rich ITO surface. This greatly affects the formation of a densely packed SAM on ITO, which is essential for efficient interfacial hole extraction. To circumvent this problem and improve the dispersion of SAM molecules in processing solvents, we noticed that the CbzNaph, MeO-2PACz, and 2PACz exhibited much greater solubility in DMF than in alcoholic solvents (≈1 mg mL^{-1} in IPA vs ≈50 mg mL^{-1} in DMF) due to stronger solvent-solute interactions. The DLS measurement and light scattering effect observation further proved that these SAMs could all be well dissolved in DMF (Figure S3, Supporting Information). Therefore, DMF could be utilized as a co-solvent to improve the dispersity of SAM molecules in solution. A small amount of DMF was utilized first to dissolve CbzNaph, MeO-2PACz, and 2PACz to obtain 50 mg mL^{-1} concentrated solutions, which were then diluted with IPA to reach the optimal processing concentrations (1 mg mL^{-1} for 2% v/v DMF in IPA, and 1.5 mg mL^{-1} for 3.3% v/v DMF in IPA).

As shown in Figure 1e, the hydrodynamic diameter of CbzNaph, MeO-2PACz, and 2PACz particles under the optimal processing concentrations in IPA (labeled as IPA-CbzNaph, IPA-MeO-2PACz, and IPA-2PACz) was 83.66, 90.37, and 68.64 nm, respectively. The transmission electron microscopy visually showed all the SAMs assembled into spherical micelles with an average diameter of 80.93 nm for IPA-CbzNaph, 87.53 nm for IPA-MeO-2PACz and 47.94 nm for IPA-2PACz (Figure S4, Supporting Information).

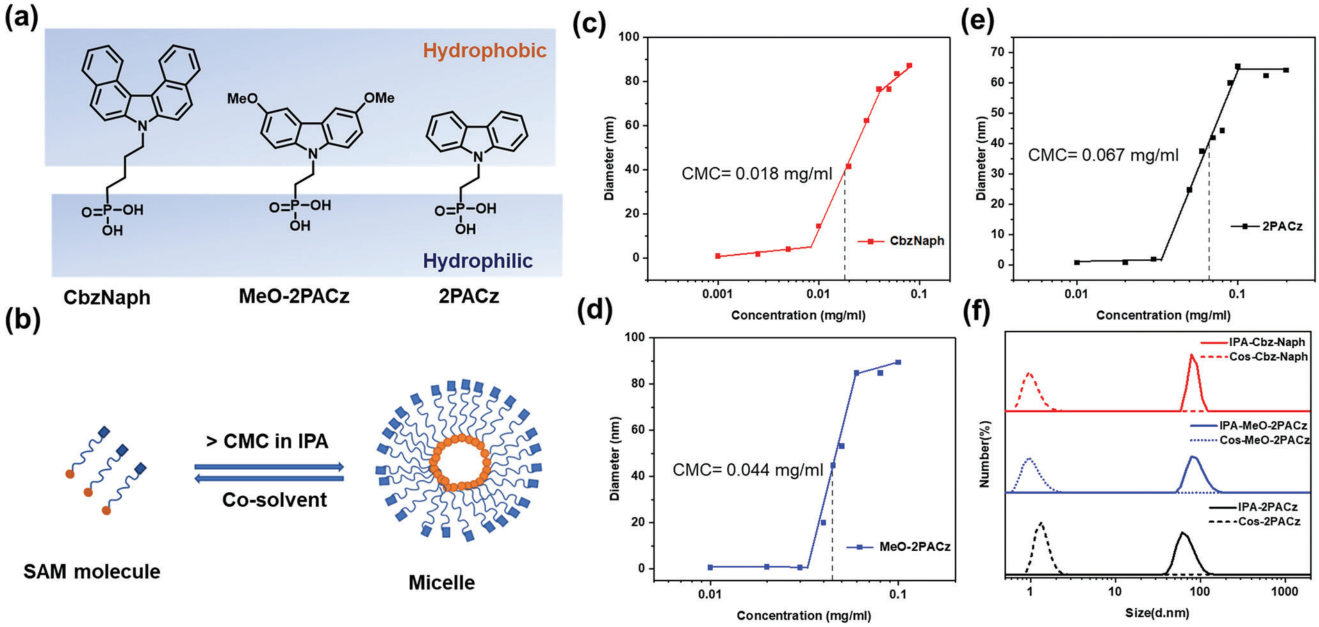

Figure 1. a) Chemical structures of CbzNaph, MeO-2PACz, and 2PACz. b) Illustration of micelles formed from the amphiphilic SAM molecules and disassembled in co-solvent. c–e) Hydrodynamic diameters in different concentrations of CbzNaph, MeO-2PACz, and 2PACz in IPA. f) Particle size distribution in solutions of IPA-SAMs and Cos-SAMs as determined by DLS.

However, it could be significantly reduced to below 1 nm in a co-solvent of 2% or 3.3% v/v DMF in IPA (labeled as Cos-CbzNaph, Cos-MeO-2PACz, and Cos-2PACz). The Tyndall light scattering effect produced by the nanoparticles formed in the solutions also disappeared, thus providing direct visual evidence of the disassembly of micelles (Figure S5, Supporting Information). The higher dielectric constant and dipole of DMF molecules can facilitate almost 8 times higher carbazole solubility than in IPA[31] to help disassemble the SAM micelles. In addition, these carbazole-based SAMs are highly soluble in aprotic solvents with high dielectric constants, such as dimethyl sulfoxide, *N*, *N*-dimethylacetamide, and *N*-methyl-2-pyrrolidone, which are also valid candidates to serve as co-solvents (Figure S6, Supporting Information).

The SAMs were formed by spin-coating from their corresponding phosphonic acids in IPA or co-solvent onto ITO substrates and then annealed at 100 °C for 10 min. Through the reaction between phosphonic acids and hydroxyl groups on ITO surface, the SAM molecules could be covalently anchored onto ITO and the residual un-bonded molecules could be removed by a washing step (Figure 2a). To assess the packing density of formed SAM, the water contact angle of ITO substrates modified with CbzNaph, MeO-2PACz, and 2PACz SAMs processed from different solvents were measured. Compared to IPA alone, when processed from the co-solvent the contact angle increased from 81.3° to 87.2° for CbzNaph, 71.7° to 77.2° for MeO-2PACz, and 74.5° to 83.4° for 2PACz (Figure 2b; Figure S7, Supporting Information), respectively. The time-dependent contact angle decay curve was also fitted, where the exponential fitting of the decay constant increased for Cos-SAMs (Figure 2c; Figure S7, Equation S1, and Table S1, Supporting Information). The initial increase and slower decay in the contact angle validate that the co-solvent strategy can universally help form denser and more regular SAMs.

Furthermore, X-ray photoelectron spectroscopy was utilized to probe the elemental contents for SAM-modified substrates within the detection depth of ≈5 nm. The presence of SAMs was proven by the characteristic signals of C 1s, N 1s, and P 2p (Figures S8–S10, Supporting Information) and the P signals of energy-dispersive X-ray spectroscopy mapping (Figure S11, Supporting Information). The surface density of SAM molecules on ITO also can be determined from cyclic voltammetry using the equation below[36]:

$$i_p = \frac{n^2 F^2}{4RTN_A} A\Gamma * \mathrm{v} \tag{1}$$

where i_p is the oxidative peak current (A), v is the voltage scan rate (V/s), n is the number of electrons transferred, the Faraday constant F = 96 485.33 C mol^{-1}, the universal gas constant R = 8.3144 J K mol, T (K) the temperature, the Avogadro constant $N_\mathrm{A} = 6.022 \times 10^{23}$ mol^{-1}, the electrode surface area A = 1.5 cm^2 in this work, and Γ* (molecules cm^{-2}) is the surface density that can be obtained by calculating the slope of i_p versus v. While being processed from the co-solvent, the surface density (Γ*) showed an increase from 2.48 × 10^{13} to 3.84 × 10^{13} molecules cm^{-2} for CbzNaph, 2.68 × 10^{13} to 3.69 × 10^{13} molecules cm^{-2} for MeO-2PACz, and 2.37 × 10^{13} to 4.34 × 10^{13} molecules cm^{-2} for 2PACz, implying denser ITO surface coverage that should correspond to improved SAM packing density (Figures S12–S14, Supporting Information).

The work function (WF) of SAMs-modified ITO was measured by ultraviolet photoelectron spectroscopy (UPS). The improved packing quality of Cos-SAMs could be corroborated by the deeper WF of modified substrates. As shown from the UPS measurements (Figure 2d; Figure S15, Supporting Information), the WF of bare ITO is 4.71 eV. The WFs of Cos-CbzNaph-,

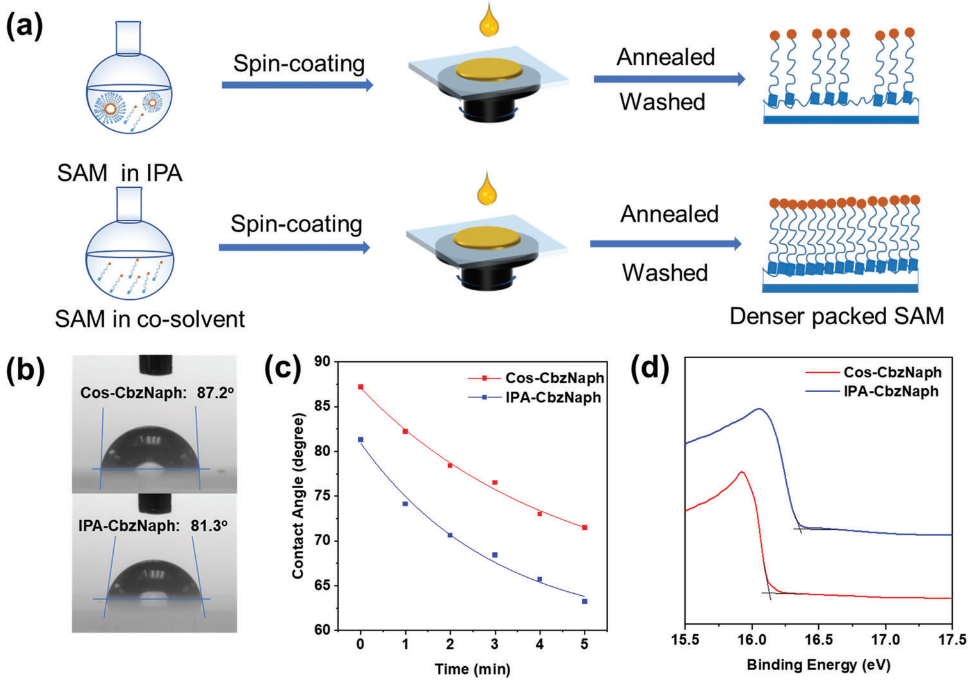

Figure 2. a) Schematic diagram for SAMs deposited from IPA or co-solvent. b,c) Water contact angles and time-dependent contact-angle evolution of ITO substrates modified with Cos-CbzNaph and IPA-CbzNaph. d) UPS spectra Cos-CbzNaph- and IPA-CbzNaph-modified ITO.

Cos-MeO-2PACz-, and Cos-2PACz-modified ITO were 5.09, 4.95, and 5.28 eV, respectively, and are deeper compared to those modified with IPA-CbzNaph (4.90 eV), IPA-MeO-2PACz (4.84 eV), and IPA-2PACz (4.91 eV). The deeper WF will help achieve better energy alignment at the perovskite/HSL interface to facilitate hole extraction, which is essential for achieving higher FF and V_{OC} of PSCs.

The photoluminescence (PL) intensity of carbazole-containing chromophores can also be used to measure the dynamics of aggregation-induced quenching in the condensed phase.[37] Using in situ PL spectroscopy to monitor the intensity change of PL during the spin-coating and thermal annealing processes allows us to understand how the co-solvent affects the kinetics of SAM growth. The CbzNaph was selected for PL measurements due to its extended π-conjugation affording a red-shifted PL peak at ≈400 nm compared to 2PACz and MeO-2PACz, which is easier to detect by our instrumental setup (Figure S16, Supporting Information). As shown in Figure 3a,c, IPA-CbzNaph showed PL quenching at ≈25 s during the spin-coating process, which can be attributed to film drying-induced aggregation. However, the PL intensity showed a three-stage evolution for Cos-CbzNaph (Figure 3b,c). The initial IPA volatilization induces molecular aggregation. Then, the slower volatilization and better solubility of CbzNaph in DMF cause the re-dissolution of CbzNaph, which increases the PL intensity. Finally, a longer PL quenching time of ≈40 s is observed due to the longer time needed to dry higher boiling point DMF. This extended phase-transition time should allow longer liquid–solid interface reaction time between the SAM and ITO, compared to the solid–solid (micelle-substrate) interface reaction in the IPA-SAM samples.

Later, during thermal annealing (100 °C for 10 min), the PL intensity of the Cos-CbzNaph decreases within the first 8 s and then stabilized due to further evaporation of DMF, indicating thermal annealing does not really affect the packing of molecules (Figure 3d). However, for the IPA-CbzNaph, a nearly three-time higher PL intensity could be observed during the period of 26 and 600 s (Figure 3e,f). As the number of anchored molecules increased during heating, the structure of SAM micelles was destabilized and the head groups no longer aggregated inwardly to the micelle core, therefore, increasing the PL intensity (Figure 3g). This additional micelle destabilization step during the SAM growing process implies extra energy is needed to generate free SAM molecules. Furthermore, the more pronounced PL intensity after thermal annealing also validated the less ordered packing of IPA-SAMs. Thus, by disassembling the SAM micelles in a proper co-solvent solution at the beginning, the energetic barrier for SAM growth caused by opening the micelles could be eliminated, which is essential for forming more densely packed SAMs with fewer defects.

Based on the above encouraging findings, inverted PSCs with a device configuration of ITO/SAM/perovskite/C60/BCP/Ag were fabricated to verify the advantage of using a co-solvent strategy to improve photovoltaic performance (Figure S17, Supporting Information) and uniform perovskite films could be deposited on all the SAM substrates (Figure S18, Supporting Information). The reference devices based on IPA-2PACz showed a maximum PCE of 21.59% with a V_{OC} of 1.135 V, a J_{SC} of 23.90 mA cm^{-2}, and an FF of 79.61%. However, the best PCE of devices based on Cos-2PACz could be improved to 22.93%, due to enhanced V_{OC} (1.167 V) and FF (82.58%) (Figure4a; Figure S19, Supporting Information). A similar trend was also found for devices based on MeO-2PACz, where the co-solvent strategy enabled an improved PCE from 21.57% to 22.89% with enhanced V_{OC} from 1.108 to 1.134 V and FF from 81.68% to 84.86% (Figure 4b;

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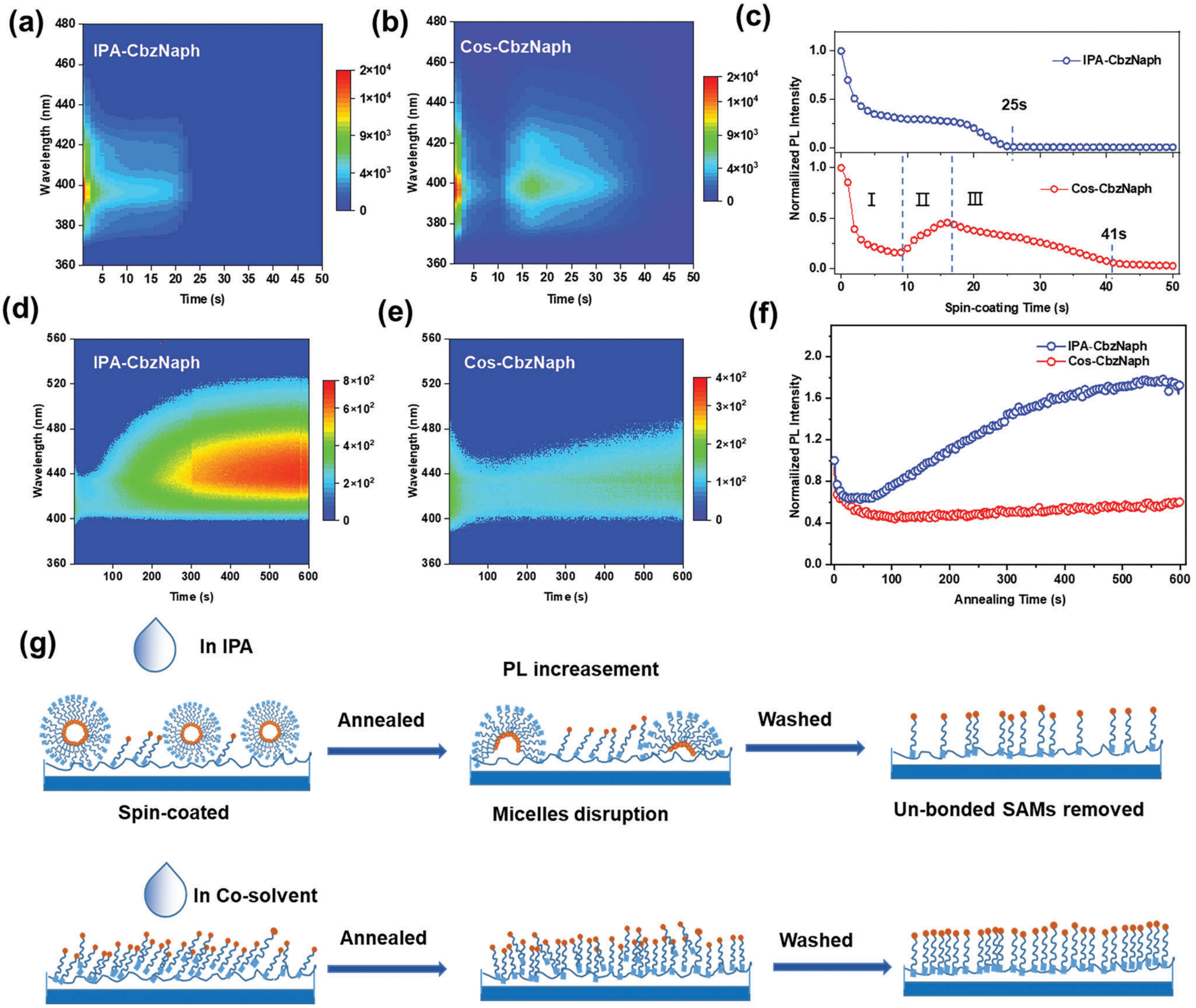

Figure 3. a,b) In situ PL intensity evolution of IPA-CbzNaph and Cos-CbzNaph in the spin-coating process. c) Evolution of the PL peak intensity over spin-coating time. d,e) In situ PL intensity evolution of IPA-CbzNaph and Cos-CbzNaph in the thermal annealing process. f) Evolution of the PL peak intensity over thermal annealing time. g) Schematic diagram for evolution of SAMs configuration adjustment deposited from IPA or co-solvent.

Figure S20, Supporting Information). As for CbzNaph, the reference device showed a maximum PCE of 22.73% with a V_{OC} of 1.152 V, a J_{SC} of 23.68 mA cm^{-2}, and an FF of 83.32%, while the Cos-CbzNaph gave a PCE of 24.02% with an improved V_{OC} (1.170 V) and very high FF (86.26%). (Figure 4c; Figure S21, Supporting Information). The external quantum efficiency (EQE) of the devices was also measured to ensure data reliability, where the results agreed well with the value extracted from the J–V curve (Figure S22, Supporting Information). These results show that the co-solvent strategy can universally improve the performance of SAM HSL-based devices. The significantly enhanced V_{OC} and FF should be due to the improved hole-extraction ability of HSL as a result of the denser SAM packing and the more down-shifted WF of ITO, forming better interfacial energy alignment as previously discussed. Moreover, the IPA-CbzNaph solution turned cloudy after being stored for 30 days, where precipitates could also be observed in the IPA-MeO-2PACz solution due to the aggregation of micelles (Figure S23, Supporting Information). In contrast, the Cos-SAM solutions remained to be clear and showed no sign of the Tyndall effect, indicating prolonged shelf-life could be achieved due to better dispersity. The devices based on the HSLs processed from aged IPA-MeO-2PACz and aged IPA-CbzNaph inks showed a lower maximum PCE of 20.53% and 21.55%, respectively. However, the aged Cos-SAM inks enabled the maintained champion PCE of 23.00% for Cos-MeO-2PACz and 24.19% for Cos-CbzNaph (Figure S24 and Table S5, Supporting Information). This shows that our co-solvent strategy can help improve reproducibility by providing a wider processing window, which is critical for the industrialization of printable PSCs.

To fully exploit the potential of the Cos-SAM HSL, light management utilizing anti-reflection coating was applied to the

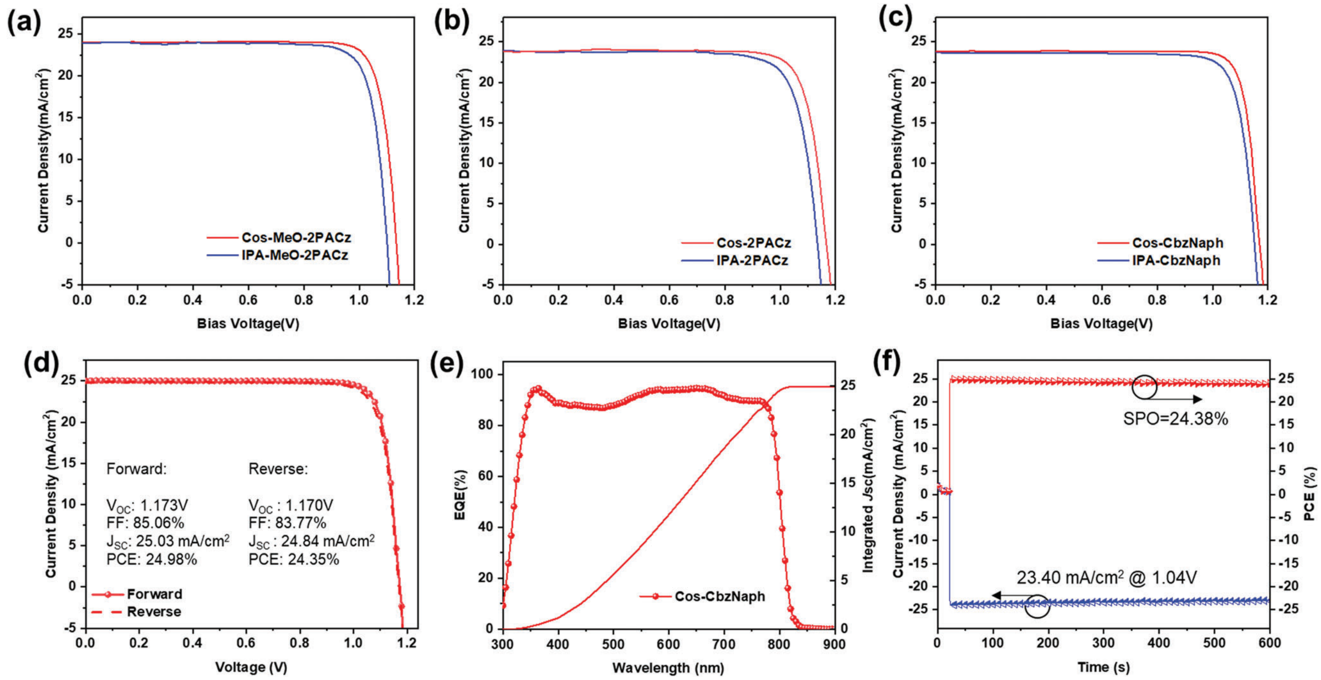

Figure 4. Current density–voltage (*J–V*) curves of PSCs with a) IPA-2PACz and Cos-2PACz; b) IPA-MeO-2PACz and Cos-MeO-2PACz; c) IPA-CbzNaph and Cos-CbzNaph HSLs. d–f) Device characteristics include *J–V* sweeps, EQE, and SPO of the champion device employing Cos-CbzNaph HSL with an anti-reflection coating.

best-performing devices based on Cos-CbzNaph. The champion PCE was improved to 24.98% (V_{OC} = 1.173 V, J_{SC} = 25.03 mA cm^{-2}, and FF = 85.06%) (Figure 4d), which is among the highest PCEs of SAM-based inverted PSCs. TheEQE of the device was also measured to ensure data reliability. As shown in Figure 4e, the integrated J_{SC} obtained from the EQE spectrum was 24.99 mA cm^{-2}, agreeing well with the value extracted from the *J–V* curve. In addition, the stabilized power output (SPO) is tracked at the maximum power point (MPP) and showed a stabilized PCE of 24.34% (Figure 4f).

The improved device performance should be directly related to balanced charge extraction and recombination at the perovskite/SAM interface. To evaluate the hole extraction ability of SAMs, the steady-state PL and time-resolved PL (TRPL) spectra of perovskites on SAM-modified substrates were measured. The weaker PL intensity of perovskite on Cos-CbzNaph-modified ITO indicated its more efficient extraction ability (Figure 5a). Moreover, the shorter charge carrier lifetime (1298.9 ns vs 1894.8 ns) showed that Cos-CbzNaph can have faster hole extraction (Figure 5b). In addition, to determine the influence of SAM density on perovskite crystallization quality, scanning electron microscopy (SEM) and X-ray diffraction (XRD) measurements were employed to investigate the quality of perovskite films on different SAM-modified substrates. The SEM images showed both homogeneous crystallization of perovskites on Cos-CbzNaph and IPA-CbzNaph (Figure S25, Supporting Information), while the XRD showed a stronger (110) crystal face diffraction of perovskite grown on the Cos-SAM HSL (Figure 5c). The improved film quality could also reduce the interfacial trap density and alleviate charge recombination to enhance the FF and V_{OC} of devices.

To assess the defect density and extent of charge carrier recombination at the perovskite/SAM interface, space charge-limited current (SCLC), dark current density, and light intensity-dependent V_{OC} measurements were also conducted. As shown in Figure 5d, the hole-only devices with an architecture of ITO/SAM/perovskite/MoO_3/Ag were fabricated for SCLC measurements. The turning point of the Ohmic region and trap-filling region of the curve revealed the trap-filled limit voltage (V_{TFL}), which were determined to be 0.719 and 0.912 V for Cos-CbzNaph- and IPA-CbzNaph-based devices, respectively. The lower V_{TFL} showed the lower trap density of devices based on Cos-SAM HSL as the trap density is directly proportional to the V_{TFL}. In addition, the reduced leakage current density was determined by the dark current density curve (Figure 5e). The V_{OC}s of the devices as a function of the incident light intensity (*P*) were also measured to calculate the ideality factor (*n*).[38] The *n* values were determined as 1.40 for Cos-CbzNaph- and 1.57 for IPA-CbzNaph-based PSCs (Figure 5f). The smaller leakage current density and *n* values demonstrated the faster hole extraction and suppressed charge recombination, which stemmed from the denser and more ordered SAM packing.

The photostability and thermal stability of PSCs based on IPA-CbzNaph and Cos-CbzNaph HSLs were also evaluated. The MPP tracking was conducted on unencapsulated devices under a N_2 atmosphere. After 700 h MPP tracking, the device with Cos-CbzNaph HSL showed better photostability, retaining 82% of its initial PCE which is higher than that of the IPA-CbzNaph-based device (70%) (Figure S26a, Supporting Information). The thermal stability of devices was also evaluated under an annealing temperature of 65 °C in a N_2-filled glove box. The device with Cos-CbzNaph HSL also exhibited better thermal stability, retaining

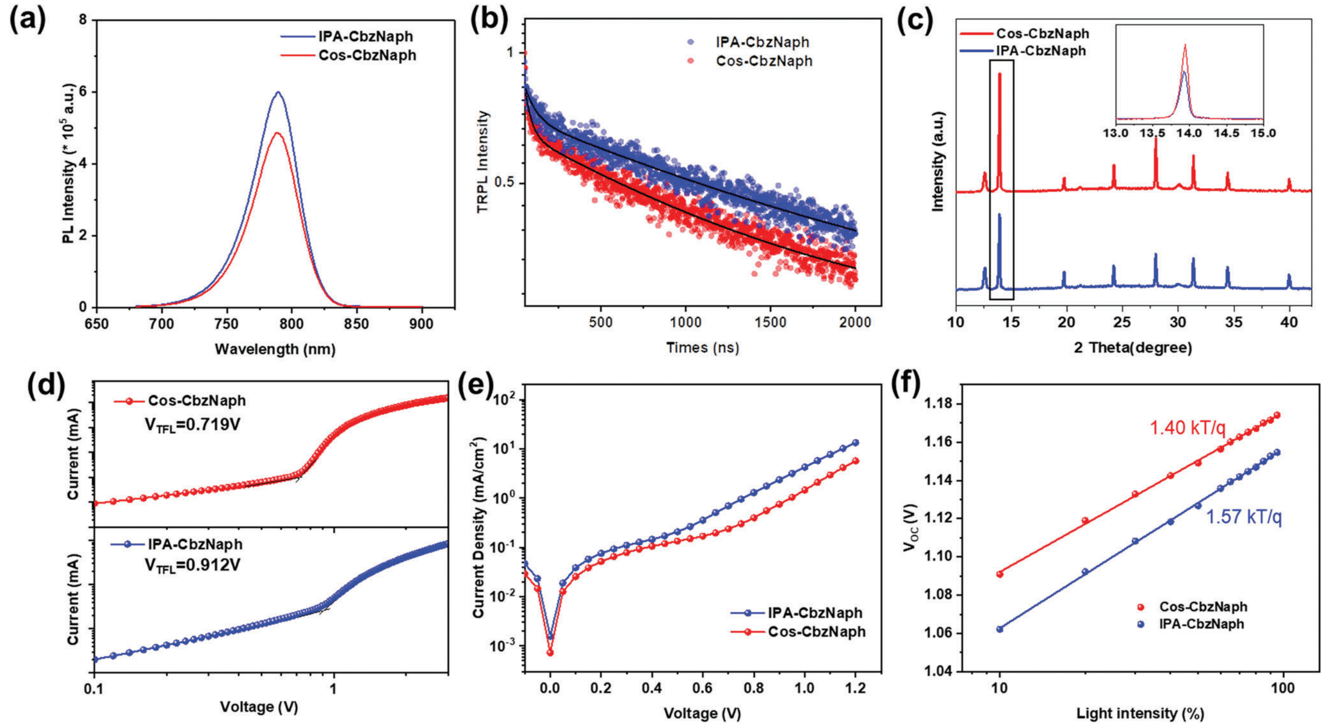

Figure 5. a) PL spectra, b) TRPL spectra, and c) the XRD patterns (inset is the (110) peaks) of perovskite films deposited on IPA-CbzNaph and Cos-CbzNaph HSL. d) SCLC of the hole-only devices with different substrates. e,f) Dark *J–V* curves and light intensity dependence of V_{OC} of PSCs with IPA-CbzNaph and Cos-CbzNaph HSL.

89% of its original PCE under continuous heating for more than 400 h, which is higher than the device with IPA-CbzNaph (81%) (Figure S26b, Supporting Information). The improved device stability could be attributed to the more ordered and densely packed Cos-SAMs, which prohibited the direct contact of perovskite with ITO to reduce interfacial defects.

3. Conclusion

In conclusion, a novel co-solvent strategy was introduced to manipulate the self-assembly of SAMs in solution to enhance their solution shelf-life, device efficiency, and stability. By introducing DMF as a co-solvent in IPA, the micelles formed by the SAM molecules are disassembled at the processing concentration. In addition, the first example of utilizing the in situ PL to understand the SAM growth process on substrates was demonstrated, where the co-solvent prolonged the liquid–solid reaction time between the phosphonic anchor group and the hydroxyl group on ITO. Moreover, the pre-disassembled micelles via co-solvent could mitigate the disruption during thermal annealing by reducing the energetic barrier to fine-tune the SAM growth process. Therefore, more ordered and denser SAMs were formed on ITO with reduced trap states, enabling better energy alignment and suppressed charge recombination at the perovskite/SAM interface. As a result, this strategy enhanced the PCEs of devices based on MeO-2PACz, 2PACz, and CbzNaph SAMs universally. The champion device based on the Cos-CbzNaph achieved a very high PCE of 24.98% with an impressive FF of 85.06% and improved stability. This work provides solid evidence on the micelle formation of carbazole-based amphiphilic SAMs in alcoholic solution and showed an effective strategy to modulate the aggregation state in solution for controlling SAM growing process. These precious insights should facilitate the development of functional SAMs for more efficient PSCs.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

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Keywords

amphiphilic molecules, carbazole, hole-selective layer, perovskite solar cells, self-assembled monolayer

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